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Comprehensive Adult Medical Eye Evaluation Preferred Practice Pattern®

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Preferred Practice Pattern® guidelines are developed by the Academy's H. Dunbar Hoskins Jr., MD Center for Quality Eye Care without any external financial support. Authors and reviewers of the guidelines are volunteers and do not receive any financial compensation for their contributions to the documents. The guidelines are externally reviewed by experts and stakeholders before publication.

COMPREHENSIVE ADULT MEDICAL EYE EVALUATION PREFERRED PRACTICE PATTERN® DEVELOPMENT PROCESS AND PARTICIPANTS

The **Preferred Practice Patterns Committee** members wrote the Comprehensive Adult Medical Eye Evaluation Preferred Practice Pattern guidelines (PPP). The committee members discussed and reviewed successive drafts of the document, meeting in person once and conducting other review by e-mail discussion, to develop a consensus over the final version of the document.

Preferred Practice Patterns Committee 2025
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The Comprehensive Adult Medical Eye Evaluation PPP was sent for review in July 2025 to improve the quality of the guideline, to gather feedback on the draft recommendations, and to assess feasibility for and applicability to the target audience, including assessing the facilitators and barriers to implementing recommendations (e.g., U.S. ophthalmologists and other important groups, including patients, other physicians, international ophthalmologists, research organizations, ophthalmological organizations, and experts in the field). The PPP was sent for review to the following patient organizations to solicit the views and preferences of patients and the public: Consumers United for Evidence-Based Healthcare, American Foundation for the Blind, Foundation Fighting Blindness, Lighthouse Guild, National Federation of the Blind, and Prevent Blindness. All those who were returning comments were required to provide disclosure of relevant relationships with industry to have their comments considered (indicated with an asterisk below). Members of the Preferred Practice Patterns Committee reviewed these comments and determined revisions to the document.

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Council*
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Practicing Ophthalmologists Advisory Committee for Education*

Invited Reviewers
American Association for Pediatric Ophthalmology and Strabismus
American Association of Ophthalmic Oncologists and Pathologists
American College of Surgeons, Advisory Council for Ophthalmic Surgery
American Foundation for the Blind*
American Glaucoma Society*
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American Society of Ophthalmic Plastic and Reconstructive Surgery
American Society of Ophthalmic Trauma
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This guideline will be formally re-evaluated and updated on a 5-year cycle in 2030. A Summary Benchmark is a resource to facilitate application of the guideline and to provide criteria that could be used to measure the application of recommendations, which will be available to all at www.aao.org/ppp.

FINANCIAL DISCLOSURES

There is no external funding, including industry/commercial support, for the development of this PPP or for the distribution of the guidelines. The Academy has fully funded the development of this PPP, and the views or interests of the Academy have not influenced the final recommendations, which are based on evidence from systematic reviews. All those individuals significantly involved in the guideline development process, including PPP Committee members, Secretary for Quality of Care, and Academy staff, have declared competing/financial interests through a financial interest disclosure process as well as an assessment of the Open Payments website (available at https://openpaymentsdata.cms.gov/). The interests of the PPP Committee members are provided at the beginning of each meeting and those with competing interests in a guideline topic do not participate in voting on areas of disagreement. In compliance with the Council of Medical Specialty Societies' Code for Interactions with Companies (available at https://cmss.org/code-for-interactions-with-companies/), relevant relationships with industry are listed. As per CMSS code, direct financial relationships with companies do not include food and beverage, research funds paid to the institution, and relationships outside of the topic of the PPP. The Academy has Relationship with Industry Procedures to comply with the Code (available at www.aao.org/about-preferred-practice-patterns). A majority (70%) of the members of the PPP Committee 2025 had no related financial relationship to disclose.

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The disclosures of relevant relationships to industry of other reviewers of the document from January to October 2025 are available online at www.aao.org/ppp.

TABLE OF CONTENTS

OBJECTIVES OF PREFERRED PRACTICE PATTERN GUIDELINES P208
METHODS AND KEY TO RATINGS P209
HIGHLIGHTED RECOMMENDATIONS FOR CARE P211
INTRODUCTION P212
Patient Population P212
Clinical Objectives P212
BACKGROUND P212
Rationale for Comprehensive Medical Eye Evaluations P212
Ocular Diseases P213
- Open-Angle Glaucoma P214
- Primary Angle-Closure Disease P215
- Diabetes Mellitus P215
- Age-Related Macular Degeneration P216
- Cataract P216
- Uncorrected Refractive Error P217
- High Myopia P217
- Other Ocular Disorders P217

Systemic Diseases and Conditions P217
Socioeconomic Considerations P217
CARE PROCESS P218
History P218
Ocular Examination P218
Diagnosis and Management P220
- Category I: Patients With No Risk Factors P220
- Category II: Patients With Risk Factors P221
- Category III: Conditions That Require Intervention P221

Clinician and Setting P222
APPENDIX 1. QUALITY OF OPHTHALMIC CARE CORE CRITERIA P223
APPENDIX 2. LITERATURE SEARCHES FOR THIS PPP P225
RELATED ACADEMY MATERIALS P227
REFERENCES P228

OBJECTIVES OF PREFERRED PRACTICE PATTERN® GUIDELINES

As a service to its members and the public, the American Academy of Ophthalmology has developed a series of Preferred Practice Pattern guidelines that **identify characteristics and components of quality eye care**. Appendix 1 describes the core criteria of quality eye care.

The Preferred Practice Pattern guidelines are based on the best available scientific data as interpreted by panels of knowledgeable health professionals. In some instances, such as when results of carefully conducted clinical trials are available, the data are particularly persuasive and provide clear guidance. In other instances, the panels have to rely on their collective judgment and evaluation of available evidence.

These documents provide guidance for the pattern of practice, not for the care of a particular individual. While they should generally meet the needs of most patients, they cannot possibly best meet the needs of all patients. Adherence to these PPPs will not ensure a successful outcome in every situation. These practice patterns should not be deemed inclusive of all proper methods of care or exclusive of other methods of care reasonably directed at obtaining the best results. It may be necessary to approach different patients' needs in different ways. The physician must make the ultimate judgment about the propriety of the care of a particular patient in light of all of the circumstances presented by that patient. The American Academy of Ophthalmology is available to assist members in resolving ethical dilemmas that arise in the course of ophthalmic practice.

Preferred Practice Pattern guidelines are not medical standards to be adhered to in all individual situations. The Academy specifically disclaims any and all liability for injury or other damages of any kind, from negligence or otherwise, for any and all claims that may arise out of the use of any recommendations or other information contained herein.

References to certain drugs, instruments, and other products are made for illustrative purposes only and are not intended to constitute an endorsement of such. Such material may include information on applications that are not considered community standard, that reflect indications not included in approved U.S. Food and Drug Administration (FDA) labeling, or that are approved for use only in restricted research settings. The FDA has stated that it is the responsibility of the physician to determine the FDA status of each drug or device he or she wishes to use, and to use them with appropriate patient consent in compliance with applicable law.

Innovation in medicine is essential to ensure the future health of the American public, and the Academy encourages the development of new diagnostic and therapeutic methods that will improve eye care. It is essential to recognize that true medical excellence is achieved only when the patients' needs are the foremost consideration.

All Preferred Practice Pattern guidelines are reviewed by their parent panel annually or earlier if developments warrant and updated accordingly. To ensure that all PPPs are current, each is valid for 5 years from the "approved by" date unless superseded by a revision. Preferred Practice Pattern guidelines are funded by the Academy without commercial support. Authors and reviewers of PPPs are volunteers and do not receive any financial compensation for their contributions to the documents. The PPPs are externally reviewed by experts and stakeholders, including consumer representatives, before publication. The PPPs are developed in compliance with the Council of Medical Specialty Societies' Code for Interactions with Companies. The Academy has Relationship with Industry Procedures (available at www.aao.org/about-preferred-practice-patterns) to comply with the Code.

The intended users of the Comprehensive Medical Adult Eye Evaluation PPP are ophthalmologists.

METHODS AND KEY TO RATINGS

Preferred Practice Pattern guidelines should be clinically relevant and specific enough to provide useful information to practitioners. Where evidence exists to support a recommendation for care, the recommendation should be given an explicit rating that shows the strength of evidence. To accomplish these aims, methods from the Scottish Intercollegiate Guideline Network[1] (SIGN) and the Grading of Recommendations Assessment, Development and Evaluation[2] (GRADE) group are used. GRADE is a systematic approach to grading the strength of the total body of evidence that is available to support recommendations on a specific clinical management issue. Organizations that have adopted GRADE include SIGN, the World Health Organization, the Agency for Healthcare Research and Policy, and the American College of Physicians.[3]

- All studies used to form a recommendation for care are graded for strength of evidence individually, and that grade is listed with the study citation.
- To rate individual studies, a scale based on SIGN[1] is used. The definitions and levels of evidence to rate individual studies are as follows:

I++	High-quality meta-analyses, systematic reviews of randomized controlled trials (RCTs), or RCTs with a very low risk of bias
I+	Well-conducted meta-analyses, systematic reviews of RCTs, or RCTs with a low risk of bias
I-	Meta-analyses, systematic reviews of RCTs, or RCTs with a high risk of bias
II++	High-quality systematic reviews of case control or cohort studies High-quality case-control or cohort studies with a very low risk of confounding or bias and a high probability that the relationship is causal
II+	Well-conducted case-control or cohort studies with a low risk of confounding or bias and a moderate probability that the relationship is causal
II-	Case-control or cohort studies with a high risk of confounding or bias and a significant risk that the relationship is not causal
III	Non-analytic studies (e.g., case reports, case series)

- Recommendations for care are formed based on the body of the evidence. The body of evidence quality ratings are defined by GRADE[2] as follows:

Good quality	Further research is very unlikely to change our confidence in the estimate of effect
Moderate quality	Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate
Insufficient quality	Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate Any estimate of effect is very uncertain

- Key recommendations for care are defined by GRADE[2] as follows:

Strong recommendation	Used when the desirable effects of an intervention clearly outweigh the undesirable effects or clearly do not
Discretionary recommendation	Used when the trade-offs are less certain—either because of low-quality evidence or because evidence suggests that desirable and undesirable effects are closely balanced

- The Highlighted Recommendations for Care section lists points determined by the PPP Committee to be of particular importance to vision and quality-of-life outcomes.
- All recommendations for care in this PPP were rated using the system described above. Ratings are embedded throughout the PPP main text in italics.
- Literature searches to update the PPP were undertaken on March 4, 2025 in the PubMed database. Complete details of the literature searches are available in Appendix 2.

- Recommendations are based on systematic reviews, as per the Institute of Medicine (Clinical Practice Guidelines We Can Trust, 2011). In formulating the recommendations, the health benefits, side effects/harms/risks, and the balance of benefits and risks are reviewed and considered. Final decisions are arrived at through informal consensus techniques. If there are areas of disagreement, a vote will be conducted among the members of the guideline committee. If there are individuals with direct financial relationships in the area of disagreement, these individuals will refrain from the vote.

HIGHLIGHTED RECOMMENDATIONS FOR CARE

A comprehensive medical eye evaluation can detect abnormalities of the visual system and uncover evidence of systemic disease with ophthalmic manifestations.

The recommended frequency for adult comprehensive medical eye examinations for asymptomatic patients without risk factors for eye disease, and for patients who do not have risk factors for eye disease, is as follows: under 40 years—every 5 to 10 years; 40 to 54 years—every 2 to 4 years; 55 to 64 years—every 1 to 3 years; and 65 years or older—every 1 to 2 years.

An increased frequency of comprehensive medical eye evaluations is recommended for adults who have risk factors for primary open-angle glaucoma, such as diabetes mellitus, older age, family history, Black race and Latino/Hispanic ethnicity.

The timing and frequency of examination for patients who have diabetes mellitus depend on the type of diabetes and pregnancy status. The recommendations are as follows: (1) type 1 diabetes mellitus—first examination 5 years after onset and yearly thereafter; (2) type 2 diabetes mellitus—first examination at the time of diagnosis and at least yearly thereafter; and (3) women who are pregnant or planning to become pregnant with type 1 or type 2 diabetes—first examination prior to conception and then early in the first trimester of pregnancy; recommendations thereafter are based on examination findings.

Smoking is a risk factor for many ocular diseases, and smoking cessation should be encouraged.

INTRODUCTION

PATIENT POPULATION

Adults with no known ocular conditions or risk factors, adults with previously identified conditions or risk factors, or adults with recurrent or new symptoms.

CLINICAL OBJECTIVES

- Detect and diagnose ocular abnormalities and diseases
- Identify risk factors for ocular disease
- Identify risk factors for systemic disease based on ocular findings
- Establish the presence or absence of ocular signs or symptoms of systemic disease
- Determine the refractive and health status of the eye, visual system, and related structures
- Discuss the results and implications of the examination with the patient
- Initiate an appropriate management plan, including determination of the frequency of future visits, further diagnostic tests, referral, or treatment

BACKGROUND

Patients may seek a comprehensive adult medical eye evaluation for a variety of reasons. It is recommended for patients who have not been examined for an extended period of time by an ophthalmologist or who are being seen for the first time. Recommended intervals between comprehensive examinations vary with age and risk factors. A thorough ophthalmic evaluation can detect common abnormalities of the visual system and related structures as well as less common yet extremely serious ones, such as ocular tumors. Such an evaluation can also uncover evidence of systemic disease that has associated ophthalmic manifestations. All patients, particularly those with risk factors for ocular disease, should be re-examined periodically to prevent or minimize vision loss by detecting and treating such diseases at an early stage. Patients in whom ophthalmic disease(s) are identified require periodic comprehensive examinations for optimal monitoring and treatment of the condition(s). With appropriate and timely intervention, potentially blinding diseases such as glaucoma, cataract, age-related macular degeneration (AMD), diabetic retinopathy, and complications of high myopia often have a more favorable outcome. Studies have indicated that up to 40% of legal blindness found among nursing home residents,[4, 5] as well as in both urban[6] and rural[7] communities, could have been prevented or ameliorated if those individuals had received timely ophthalmic screening and care. In a population-based study, 63% of Latino participants who had eye disease were not aware of it.[8]

RATIONALE FOR COMPREHENSIVE MEDICAL EYE EVALUATIONS

The rationale for performing periodic comprehensive medical eye evaluations in adults without known ocular conditions, symptoms, or risk factors is to detect ocular diseases, visual dysfunction, or ophthalmic signs of systemic disease. Early recognition, counseling, or treatment may preserve visual function or, in the case of systemic diseases, could prevent serious illness or even premature death. Irreversible vision loss has been associated with adverse effects on quality of life and mental health,[9, 10] and self-reported visual loss has been found to be significantly associated with depression.[11] Comprehensive medical eye evaluations are also performed periodically to evaluate new symptoms and monitor patients with previously identified eye conditions or risk factors.

The public health impact of eye disease is substantial, because vision affects daily functioning.[12-16] Improvement in visual function that occurs as a result of treatment of ocular disorders is accompanied by improvement in life satisfaction and mental health and by participation in home and community activities.[17-20] Vision plays a critical role in mobility and in fall prevention.[21-24] Untreated visual impairment has been associated with cognitive decline and Alzheimer‘s disease.[25] A higher risk of motor vehicle collisions was found among drivers with glaucoma who had severe visual field

defects.[26] Cataract surgery in older drivers has been shown to reduce the subsequent motor vehicle collision rate.[27] Visual impairment, AMD, and cataract have been associated with an increased risk of mortality.[28, 29] In women 65 and older, poorer visual acuity and reduced contrast sensitivity have been associated with a higher risk of mortality.[30]

OCULAR DISEASES

Many patients will be unaware that they have uncorrected refractive error or a vision-threatening ocular condition because of the lack of early symptoms (see Table 1). These conditions include common and often treatable diseases such as glaucoma, diabetic retinopathy, and some forms of macular degeneration.

TABLE 1 PREVALENCE OF OCULAR DISEASES AND CONDITIONS THAT MAY BE ASYMPTOMATIC

Disease or Condition	Prevalence	Risk Factors for Disease or Disease Progression	Potentially Positive Findings on Examination
Open-angle glaucoma	Black individuals age ≥40 years: 3.4%[31] Hispanic individuals age ≥40 years: 2%[32]–4.7%[33] White individuals age ≥40 years: 1.7%[31]	Black race or Latino/Hispanic ethnicity,[31, 33-35] older age,[31, 32, 34-37] family history of glaucoma,[38, 39] elevated IOP,[40, 41] thinner central cornea,[40, 41] type 2 diabetes mellitus,[42-45] lower systolic and diastolic blood pressure,[46, 47] lower ocular perfusion pressure,[46-49] myopia,[49-54] lower corneal hysteresis,[55-57] disc hemorrhage,[58-62] larger cup-to-disc ratio,[40, 63] higher pattern standard deviation[40, 64, 65]	Optic disc cupping and retinal nerve fiber layer defect, characteristic visual field defect, elevated IOP, decreased vision (late stages), secondary causes of IOP elevation (e.g., exfoliation material on the lens capsule, signs of pigment dispersion syndrome)
Primary angle-closure glaucoma	0.009%[66]–2.6%[67] (highest rates in Inuit and Asian populations) Hispanic individuals age >40 years: 0.1%[32]	Hyperopia, family history of angle closure, older age,[68] female sex,[69, 70] Inuit or Asian descent,[68, 71-73] and ocular biometrics (reduced angle width, decreased anterior chamber depth, shorter axial length, increased lens thickness)[74-79]	Narrow angles, elevated IOP, peripheral anterior synechiae, increased lens thickness (vaulting), plateau iris configuration, optic nerve cupping
Diabetic retinopathy	General population age ≥40 years: 3.4%[80] Individuals age ≥40 years with type 1 or type 2 diabetes: 28.5%[81]–40.3%[80] Hispanic individuals with type 1 or type 2 diabetes age ≥40 years: 46.9%[82]	Increasing duration of diabetes,[82, 83] high levels of glycosylated hemoglobin,[84-91] high systolic blood pressure,[92, 93] elevated serum lipid levels[94-96]	Retinal microaneurysms, hemorrhages, lipid exudates, intraretinal microvascular anomalies, macular edema, retinal or anterior segment neovascularization, preretinal or vitreous hemorrhage, tractional retinal detachment
Early or Intermediate AMD	General US population age ≥ 40 years: 11.6%[97] Hispanic individuals age ≥ 40 years: 12.2%[97] Non-Hispanic Black individuals age ≥ 40 years: 7.2%[97] Non-Hispanic White individuals age ≥ 40 years: 12.3%[97] Other race and ethnicity* age ≥ 40 years: 10.5%[97]	Increasing age,[98-100] bilateral soft drusen, large drusen, confluent drusen, clumping or atrophy of retinal pigment epithelium,[101-103] family history,[104, 105] genetic polymorphisms,[106-118] smoking,[119] Mediterranean diet[120-122] (protective)	Retinal pigment epithelial disturbances/atrophy, intermediate or large drusen, geographic atrophy, or retinal pigmented epithelial detachments

TABLE 1 PREVALENCE OF OCULAR DISEASES AND CONDITIONS THAT MAY BE ASYMPTOMATIC

Disease or Condition	Prevalence	Risk Factors for Disease or Disease Progression	Potentially Positive Findings on Examination
Late AMD	General US population age ≥40 years: 0.9 %[97] Hispanic individuals age ≥40: 0.4%[97] Non-Hispanic Black individuals age ≥40: 0.7%[97] Non-Hispanic White individuals age ≥40: 1.0%[97] Other race and ethnicity* age ≥40: 1.0%[97]	Increasing age,[98-100] family history, smoking, bilateral soft drusen, large drusen, confluent drusen, clumping or atrophy of retinal pigment epithelium,[123] greater body mass index,[124] genetic factors,[106-114, 125] Mediterranean diet (protective)[126-131]	Drusen and associated retinal pigment epithelial changes, geographic atrophy, evidence of choroidal neovascularization (intra- or subretinal hemorrhage, lipid, intra- or subretinal fluid)
Choroidal nevi	5%–8%, increases with age, and more common in White individuals.[132] (Note: Findings are based on 45º fundus images centered on the fovea and optic nerve.)	White race and increasing age[132]	Clearly defined margins, often flat or slightly elevated; typically stable in size. Over time, choroidal nevi may display overlying drusen, retinal pigment epithelial atrophy, hyperplasia, or fibrous metaplasia.

AMD = age-related macular degeneration; IOP = intraocular pressure.

*Other race and ethnicity included Alaskan Native individuals, American Indian individuals, Asian individuals of any origin, multiracial individuals, Pacific Islander individuals, and other groups not otherwise classified.

In 2017, about 1.08 million people in the United States were legally blind (best-corrected visual acuity of ≤20/200 in the better-seeing eye), and 7.08 million people in the United States had visual acuity loss (best-corrected visual acuity of ≤20/40 in the better-seeing eye).[133] The highest frequencies of visual acuity loss and legal blindness were found in individuals 85 years and older and generally correlated with age.[134, 135] Rates of visual acuity loss and legal blindness were disproportionately higher among Hispanic and non-Hispanic Black individuals compared with non-Hispanic White individuals.[133]

There is a growing recognition that racial and ethnic categories are socially constructed and not biologically coherent, and concerns about the use of these terms in medical research have been increasingly acknowledged.[136] Many studies have reported associations between race or ethnicity and health outcomes, but outcomes may also be related to ancestry and heritage, as well as socioeconomic, structural, institutional, cultural, demographic, and other factors.[137-139] Identifying the race and ethnicity of study participants may provide information about the generalizability of results. However, many individuals identify with more than one race and ethnicity, and categories should not be viewed as absolute. Whenever possible, rather than solely relying on self-identified race and ethnicity, researchers should identify which other factors (e.g., economic, environmental, social) might be conflated with race and ethnicity and incorporate multiple types of measures into the study design. Race and ethnicity categories should be clearly and specifically defined, including how the data were collected. The use of nonspecific categories such as “Other” or “non-White” should be avoided, and participants should be able to select more than one category to account for multiracial and multiethnic individuals.[140]

Ocular disorders, including high-risk conditions and diseases, that necessitate a comprehensive medical eye evaluation include the following.

Open-Angle Glaucoma

Primary open-angle glaucoma (POAG) is a significant public health problem.[31, 33, 35, 141-146] It was estimated that 76 million people in the world had glaucoma in the year 2020.[142] Glaucoma (both open-angle and angle-closure) is the second-leading cause of blindness worldwide.[143] Overall, the prevalence of POAG worldwide among adults aged 40 to 80 was estimated to be 3.05% in 2013.[142] Prevalence studies suggest that the number of patients with POAG will increase by 50% worldwide, from 52.7 million in 2020 to 79.8 million in 2040 as the population ages,[142] and will disproportionately affect residents of African and Asian

countries.[141, 142, 144, 145] Large differences exist in the prevalence of glaucoma among different racial and ethnic groups. Overall, there appears to be a threefold higher prevalence of open-angle glaucoma (OAG) in Black individuals relative to non-Hispanic White individuals in the United States.[31, 35] It is also the leading cause of blindness in Black individuals.[35] Further, the prevalence of OAG is even higher in Afro-Caribbean individuals relative to African American individuals. Hispanic/Latino individuals have high prevalence rates of OAG that are comparable to the prevalence rates for Black individuals.[33] An analysis of claims data from a large U.S.-based managed care plan suggests that the prevalence of OAG among Asian American individuals is comparable to the prevalence among Latino individuals and is higher than that of non-Hispanic White American individuals. However, it is important to acknowledge that self-identified race and ethnicity categories used in research studies are inherently limited and intertwined with other aspects of health, such as socioeconomic, structural, institutional, cultural, and demographic factors.[140]

Primary Angle-Closure Disease

Although POAG is approximately twice as common as primary angle-closure glaucoma (PACG), PACG confers a twofold higher risk of blindness.[147] Considerable differences exist in the prevalence of angle closure among racial and ethnic groups, with highest rates in Inuit,[68, 148-151] Chinese,[67, 152, 153] and other Asian populations[154-162]; lower rates are reported in populations of African origin[35, 163, 164] and European origin.[66, 165-168] In some Asian populations, PACG is reported to account for nearly as many cases of glaucoma as OAG.[71, 142, 143, 154, 169] Worldwide, 0.7% of people over 40 years of age are estimated to have angle-closure glaucoma;[143] in 2013, this represented 20.2 million people, with most (15.5 million) in Asia.[142] In China, PACG is estimated to cause unilateral blindness (visual acuity <20/200 or visual field ≤10º) in 1.5 million individuals and bilateral blindness in another 1.5 million.[169]

Diabetes Mellitus

An estimated 133 million Americans aged 18 years and older have been diagnosed with either diabetes or prediabetes, according to a report based on the 2020 census data by the Centers for Disease Control and Prevention (CDC).[170] Estimates for 2021 reported by the CDC were that 38.4 million Americans 18 or older had diabetes (11.6% of that age group), and 22.8% were not aware that they had the disease.[170] An additional estimated 97.6 million U.S. adults have impaired fasting blood glucose levels (based on both fasting blood glucose levels and HbA_{1c} levels).[170]

Rates of patients with diagnosed diabetes increased with age: 4.8% of individuals 18 to 44 years old had diabetes, as did 18.9% of those 45 to 64 years old and 29.2% of those 65 and older. Rates of diagnosed diabetes were higher among non-Hispanic American Indian or Alaskan Native individuals (16.0%), non-Hispanic Black individuals (12.5%), and Hispanic individuals (10.3%) compared with Asian individuals (9.2%) and non-Hispanic White individuals (8.5%).[170]

Rates of prediabetes (HbA_{1c} levels between 5.7% and 6.4%) are also increasing.[171] It is estimated that 38% of U.S. adults 18 or older (97.6 million people) have prediabetes based on their fasting glucose or HbA_{1c} level.[170] The age-adjusted prevalence of diabetes from 2019 to 2021 was higher for people with less than a high school education (13.1%) compared with those with more than a high school education (6.9%).[170] Rates of diabetes and prediabetes are similarly high among children and adolescents (under age 20) at 35 per 10,000; 86% had type 1 diabetes.[170]

According to estimates based on data from the U.S. Census Bureau, approximately one third of Americans are at risk of developing diabetes mellitus during their lifetime. With increasing industrialization and globalization, there has been a concomitant increase in the prevalence of diabetes that has led to a worldwide public health issue.[172] An alarming increase in the frequency of type 2 diabetes in the pediatric age group has been noted in several countries,[86, 173-177] including the United States, and has been associated with an increased frequency of childhood obesity.[178] In addition, there is evidence suggesting that diabetes develops at earlier ages and carries a higher incidence of complications among racial and ethnic minorities.[179-181]

Life conditions including income, housing, and education that affect health (social determinants of health) also impact the prevalence and incidence of diabetes mellitus.[182]

Overall, demographics, health care access, psychological distress, the level of education attained, and financial wealth, as well as the reported associations of race and ethnicity with health outcomes, remain strong predictors of mortality risk among adults with diabetes.[183]

Age-Related Macular Degeneration

Age-related macular degeneration (AMD) is a leading cause of severe vision impairment in developed countries.[35, 134, 184-187] In 2019, there were an estimated 20 million individuals in the United States with AMD; of these, 18.34 million had early stages of AMD and 1.49 million had late stages of the disease.[97] Although the prevalence, incidence, and progression of AMD and most associated features (e.g., large drusen) increase with age, the prevalence also varies by race and ethnicity.[134, 188-190] The prevalence is increasing, especially with an aging global population, leading to an estimated 196 million patients with AMD in 2020 and a projected 288 million by 2040.[191] Age-related macular degeneration is usually asymptomatic in its early stages, although a fundus examination helps identify patients with an increased risk of developing choroidal neovascularization or advanced AMD.[123] The original Age-Related Eye Disease Study (AREDS) results defined a role for nutritional supplements (vitamins C and E, beta-carotene, and zinc) for slowing the progression of AMD. Results of AREDS2 support the removal of beta-carotene and the addition of lutein/zeaxanthin. According to AREDS2, antioxidant vitamin and mineral supplementation should be considered in patients with intermediate AMD or geographic atrophy in one or both eyes and other advanced AMD in one eye.[192] An estimated 8 million persons at least 55 years old in the United States are at high risk for developing advanced AMD and are the population for whom the AREDS2 formulation should be considered. If all the patients at risk were given supplements, then more than 300,000 could delay disease progression and associated vision loss.

Cigarette smoking has been consistently identified in numerous studies as a risk factor for incident AMD and progression of AMD, and the risk increases relative to the number of pack-years smoked.[119, 120, 124, 193-198] Smoking-cessation counseling may influence patients to stop smoking, reducing the risk of AMD progression. Patients with neovascular AMD report a substantial decline in their quality of life and have an increased need for assistance with activities of daily living that progresses as visual acuity worsens.[199] Early treatment of AMD is associated with a more favorable prognosis.[200] Anti-vascular endothelial growth factor therapies have become the first-line therapy for treating and stabilizing most cases of neovascular AMD, and a Cochrane systematic review demonstrated the effectiveness of these agents to maintain visual acuity.[201] It is important to note that since the registration trials for the currently approved anti-vascular endothelial growth factor medications, the standard of care is to treat neovascular AMD as soon as the diagnosis has been made. Because early symptoms may be subtle, a comprehensive eye examination may represent a patient's best opportunity to be diagnosed and treated at an earlier and potentially more favorable stage.

Two therapies for non-neovascular AMD with geographic atrophy have been shown to reduce geographic atrophy growth and have received regulatory approval. The first was pegcetacoplan 15 mg/0.1ml (SYFOVRE, Apellis Pharmaceuticals, Waltham, MA), which blocks the C3 and C3b proteins of the complement-mediated immune system.[202, 203] The second approved treatment was an avacincaptad pegol intravitreal solution 2 mg (Izervay, Astellas Pharma US, Northbrook, IL), a PEGylated, stabilized aptamer that targets C5. There was no benefit for best corrected visual acuity in the clinical trials for either pegcetacoplan or avacincaptad pegol, although the post hoc analysis for avacincaptad pegol suggested a possible slowing of progression to persistent vision loss. Because of the risk of exudative AMD, patients should be monitored and treated if needed.

Cataract

Cataract remains a significant cause of visual disability in the United States, accounting for approximately 50% of low-vision cases in adults over 40.[204] Because smoking increases the risk of cataract progression,[205, 206] informing smokers about this and other associated ocular and systemic diseases may influence them to stop smoking.

Uncorrected Refractive Error

Uncorrected refractive error may be asymptomatic and disabling. It can be detected on comprehensive medical evaluation by history and visual acuity testing with pinhole; refraction can be undertaken at that time or upon referral to an appropriate clinician. Over half of Americans older than 40 have ametropia of sufficient magnitude to require refractive correction.[207] A 2021–2022 cross-sectional study report found that uncorrected refractive error was the main cause of vision impairment at two Michigan clinics serving low-income communities and was associated with reduced vision-related quality of life.[208]

High Myopia

The prevalence of myopia is increasing globally, and it is estimated that by 2050 half of the world population will be myopic and 10% will be highly myopic.[209] It is also estimated that by 2050 visual impairment due to myopic macular degeneration will grow from 10.0 million to 55.7 million people worldwide (prevalence, 0.57%; 95% confidence interval [CI], 29.0–119.7 million), and blindness will grow from 3.3 million to 18.5 million (prevalence, 0.19%; 95% CI, 9.6–39.7 million).[210] Vision loss occurs in high myopia due to myopic macular degeneration, also known as myopic maculopathy or myopic retinopathy.[211] In addition, high myopia and abnormalities of the posterior segment, such as retinal tears or retinal degenerations (i.e., lattice degeneration or subclinical asymptomatic retinal detachments), increase the risk of retinal detachment.

Other Ocular Disorders

Other examples of high-risk conditions or diseases that necessitate a comprehensive medical eye evaluation include a history of ocular trauma or the presence of abnormalities of the anterior segment, such as corneal ectasia, corneal dystrophies, or peripheral anterior synechiae. Conditions that increase the risk of OAG (e.g., exfoliation syndrome and pigment dispersion syndrome) and angle-closure glaucoma (narrow anterior chamber angle) should also be evaluated.

SYSTEMIC DISEASES AND CONDITIONS

Systemic infectious, neoplastic, autoimmune, cardiovascular, and nutrition-related diseases may be discovered during the comprehensive medical eye evaluation. The following components of the comprehensive examination may identify signs of systemic diseases or other serious medical conditions, such as:

- External examination: orbital tumor, Graves' disease, metabolic storage diseases, face/head trauma
- Pupil examination: Horner's syndrome, pharmacologic toxicity, midbrain tumor, aneurysm
- Ocular alignment and motility: neurological disorders (e.g., myasthenia gravis, central nervous system defects or aneurysm, multiple sclerosis), Graves' disease, congenital abnormalities, trauma
- Visual fields by confrontation: cerebrovascular accidents, chiasmal tumors
- Anterior segment: drug or heavy-metal toxicity; autoimmune diseases; infectious diseases; vitamin A deficiency; metabolic, endocrine, or storage diseases; chromosomal abnormalities; congenital abnormalities; sleep apnea
- Lens: Alport syndrome, Apert syndrome, atopic disease, juvenile rheumatoid arthritis, myotonic dystrophy, Wilson's disease, homocystinuria, Marfan syndrome, Weill-Marchesani syndrome, history of topical or systemic corticosteroid use
- Posterior segment: systemic hypertension; diabetes mellitus; infectious diseases (e.g., HIV, tuberculosis, syphilis, histoplasmosis, toxoplasmosis); immune-mediated diseases; vasculitis; primary or metastatic tumors; metabolic storage diseases; phakomatoses; hematologic diseases; cerebrovascular disease; increased intracranial pressure; toxicity from hydroxychloroquine, tamoxifen, or phenothiazines; trauma
- Optic nerve: malignancy, trauma, idiopathic intracranial hypertension

SOCIOECONOMIC CONSIDERATIONS

In 2012, the costs of vision loss and eye disorders among the population younger than 40 years were estimated at $27.5 billion (95% CI, $21.5–$37.2 billion), including $5.9 billion for children and $21.6

billion for adults 18 to 39 years of age in the United States. This total included $14.5 billion in direct costs: $7.3 billion for diagnosed eye disorders, $4.9 billion in refraction correction, and $0.5 billion for undiagnosed vision loss. The indirect costs were $13 billion, due mainly to productivity losses. In addition, this cumulative vision loss cost society 215,000 quality-adjusted life years.[212] There were significant differences in the use of eye care services by adults with eye diseases in the United States with respect to socioeconomic status, as measured by poverty-income ratio and educational attainment.[213] In 2017, the estimated economic burden of self-reported vision loss and blindness in the United States was $134.2 billion, $35.5 billion of which was in indirect costs and $98.7 billion of which was in direct costs.[214]

CARE PROCESS

A comprehensive medical eye evaluation includes a history, examination, diagnosis, and treatment. The evaluation includes detecting and diagnosing ophthalmic disorders, developing a treatment plan for addressing refractive error and ocular disease, and referring detected systemic disease to the appropriate medical clinician. The items listed are basic areas of evaluation or investigation and are not meant to exclude additional elements when appropriate. For example, because history taking is an interactive process, the patient's responses may guide the clinician to pursue additional questions and evaluation.

HISTORY

In general, a thorough history may include the following items:

- Demographic data (e.g., name, date of birth, gender, ethnicity, and race)
- Patient’s other pertinent health care clinicians
- Chief complaint and history of present illness
- Present status of visual function (e.g., patient’s self-assessment of visual status, visual needs, any recent or current visual symptoms, and use of eyeglasses or contact lenses)
- Ocular symptoms (e.g., eyelid swelling, diplopia, redness, photophobia)
- Ocular history (e.g., prior eye diseases; injuries; surgery, including cosmetic eyelid and refractive surgery; or other treatments and medications)
- Systemic history: medical conditions and previous surgery
- Medications: ophthalmic and systemic medications currently used, including nutritional supplements and other over-the-counter products
- Allergies or adverse reactions to medications
- Family history: pertinent familial ocular (e.g., glaucoma, AMD) and systemic disease
- Social history (e.g., occupation; tobacco, alcohol, illicit drug use; sexual activity; family and living situation, as appropriate)
- Directed review of systems

OCULAR EXAMINATION

The comprehensive medical eye evaluation consists of an evaluation of the physiological function and the anatomical status of the eye, visual system, and its related structures. This usually includes the following elements:

- Visual acuity with current correction (the power of the present correction recorded) at distance and with pinhole if reduced, and, when appropriate, at near
- Refraction when indicated
- Visual fields by confrontation
- External examination (e.g., eyelid position and character, lashes, lacrimal apparatus and tear function; globe position; and pertinent skin and facial features)
- Pupil examination (e.g., size and response to light, relative afferent pupillary defect)
- Ocular alignment (e.g., cover test) and motility (e.g., ductions and versions)

- Slit-lamp biomicroscopic examination: eyelid margins and lashes; tear film; conjunctiva; sclera; cornea; anterior chamber, including central and peripheral anterior chamber depth; iris; lens; and anterior vitreous
- Intraocular pressure measurement (using a contact applanation method is most accurate, typically a Goldmann tonometer). Contact tonometry may be deferred in the setting of suspected ocular infection or ocular trauma.
- Fundus examination: mid and posterior vitreous, retina (including posterior pole and periphery), vasculature, and optic nerve
- Assessment of relevant aspects of the patient's mental and physical status

Examination of anterior segment structures routinely involves gross and biomicroscopic evaluation before and after dilation. Evaluation of structures situated posterior to the iris is best performed through a dilated pupil. Optimal examination of the optic nerve, macula, and peripheral retina requires the use of an indirect ophthalmoscope or slit-lamp fundus biomicroscopy with appropriate accessory diagnostic lenses.

Based on the patient's history and findings, additional tests or evaluations might be undertaken to further evaluate a particular structure, function, or problem. These are not routinely part of the comprehensive medical eye clinical evaluation. Specialized clinical evaluation may include the following:

- Monocular near-vision testing
- Potential acuity testing
- Glare testing
- Contrast sensitivity testing
- Color-vision testing
- Testing of stereoacuity and fusion
- Testing of accommodation and convergence
- Central visual field testing (Amsler grid)
- Expanded evaluation of ocular motility and alignment in multiple fields of gaze at distance and near
- Exophthalmometry (i.e., Hertel)
- Tear breakup time
- Ocular surface staining (e.g. fluorescein, rose bengal)
- Corneal sensation
- Gonioscopy
- Functional evaluation of the nasolacrimal system
- Indirect ophthalmoscopy with scleral indentation
- Contact lens stereoscopic biomicroscopy (e.g., Goldmann three-mirror lens)

Additional diagnostic testing may include the following:

- Keratometry (e.g., to assess surface quality and power)
- Corneal topography/tomography, including analysis
- Measurement of corneal thickness (optical and ultrasonic pachymetry)
- Corneal endothelial cell analysis
- Meibomography
- Tear osmolarity
- External, slit-lamp, or fundus photography (e.g., wide field)
- Anterior and posterior segment optical coherence tomography
- Confocal microscopy
- Wavefront analysis
- Visual fields by automated and/or manual perimetry
- Biometry for intraocular lens calculations
- Stereophotography or computer-based image analysis of the optic disc and retinal nerve fiber layer or macula
- Ophthalmic ultrasonography (A-scan, B-scan, ultrasound biomicroscopy)
- Fluorescein, indocyanine green, and optical coherence tomography angiography

- Fundus autofluorescence
- Electrophysiological testing
- Microbiology and histology/cytology of ocular or periocular specimens
- In-office point-of-care testing (for matrix metalloproteinase -9, osmolarity, adenovirus)
- Radiologic imaging
- Laboratory tests for systemic disease or genetic disorders

DIAGNOSIS AND MANAGEMENT

The ophthalmologist evaluates and integrates the findings of the comprehensive ophthalmic examination with all aspects of the patient's health status and social situation in determining an appropriate course of action. Patients fall into 1 of 3 general categories based on the results of the evaluation: no risk factors, risk factors, and conditions that require intervention.

Category I: Patients With No Risk Factors

When the initial comprehensive evaluation is normal or involves only refractive errors that require corrective lenses, the ophthalmologist reviews the findings with the patient and renders advice regarding an appropriate interval for re-examination. Although this is considered a low-risk category, periodic re-examination is indicated to detect new, potentially asymptomatic, or unrecognized ocular disease, such as glaucoma, diabetic retinopathy, and AMD, as the incidence of these ocular diseases increase with age.

A 5-year observational study of a nationally representative cohort of Medicare beneficiaries showed that patients 65 and older who had more regular eye examinations experienced less decline in vision and functional status than those who had less frequent examinations.[215] For each additional year in which a patient received an eye examination, there was an increased likelihood of continuing to read newsprint and maintaining activities of daily living. There was also a decreased risk of developing new limitations in activities of daily living and instrumental activities of daily living. Instrumental activities of daily living are activities related to independent living and include preparing meals, managing money, shopping for groceries or personal items, performing light or heavy housework, and using a telephone.

There is little published evidence to define the optimal frequency of eye examinations of patients under 65 with no eye symptoms or signs. There is some evidence that clinically significant fundus abnormalities in asymptomatic patients increase with age,[216] but other evidence suggests that the diagnostic yield of dilated fundus examination in asymptomatic patients is low, particularly in younger age groups.[217] In the absence of symptoms or other indications following the initial comprehensive medical eye evaluation, periodic evaluations are recommended at the frequency indicated in Table 2, which takes into account the relationship between increasing age and the risk of asymptomatic or undiagnosed disease. At the time of each comprehensive medical eye evaluation, the ophthalmologist will reassess the patient to determine the appropriate follow-up interval. Adults with no signs or risk factors for eye disease should have a comprehensive medical eye evaluation at age 40 years if they have not previously received one.[218]

Interim evaluations, such as screenings, refractions, or less extensive evaluations, are indicated to address episodic minor problems and complaints, or for patient reassurance. Other situations may warrant a comprehensive medical eye evaluation. The extent of the interim evaluation to be performed is determined by the patient's condition, symptoms, and the ophthalmologist's medical judgment.

TABLE 2 COMPREHENSIVE MEDICAL EYE EVALUATION FOR ADULTS WITH NO RISK FACTORS

Age (years)	Frequency of Evaluation*
Under 40	5–10 years
40–54	Every 2–4 years
55–64	Every 1–3 years
65 or older[215]	Every 1–2 years

* Interim eye evaluations, consisting of vision examinations (e.g., refractions, eyeglasses, contact lens evaluations), may be performed during these periods as well.

Category II: Patients With Risk Factors

A patient is considered to be at increased risk when the evaluation reveals signs that are suggestive of a potentially abnormal condition or when risk factors for developing ocular disease are identified but the patient does not yet require intervention. These situations may merit closer follow-up to monitor the patient's ocular health and to detect early signs of disease with additional testing.

The ophthalmologist determines an appropriate follow-up interval for each patient based on the presence of early symptoms and signs, risk factors, the onset of ocular disease, and the potential rate of progression of a given disease. For example, patients with high myopia may require annual retinal exams. Individuals of African descent might require more frequent examinations because they are at higher risk for an earlier onset and more rapid progression of glaucoma. It is recommended that patients with the conditions and risk factors noted in Table 3 undergo a comprehensive medical eye evaluation at the listed intervals.

TABLE 3 COMPREHENSIVE MEDICAL EYE EVALUATION FOR PATIENTS WITH DIABETES MELLITUS OR RISK FACTORS FOR GLAUCOMA

Condition/Risk Factor*	**Frequency of Evaluation†**	
Diabetes Mellitus	Recommended Time of First Examination	Recommended Follow-up*
Type 1[219]	5 years after onset†	Yearly‡
Type 2[220]	At time of diagnosis	Yearly‡
Prior to pregnancy[221-223] (Type 1 or 2)	Prior to conception and early in the first trimester	See Diabetic Retinopathy PPP[224] for interval recommendations based on findings at first examination
Risk Factors for Glaucoma[31-33, 40, 41, 63]	Frequency of Evaluation*	
Age 65 years or older	Every 1–2 years*	
Age 55–64 years	Every 1–2 years	
Age 40–54 years	Every 1–3 years	
Under 40 years	Every 2–5 years	

* The Center for Medicare and Medicaid Services covers glaucoma examinations by eye care professionals in the office for beneficiaries who have diabetes mellitus, those with a family history of glaucoma, African American individuals 50 years or older, and Hispanic individuals 65 years or older.

† The ophthalmologist's assessment of degree of risk, abnormal findings, or potential loss of visual function may dictate more frequent follow-up examinations than listed in this table. If the patient has additional glaucoma risk factors, the Primary Open-Angle Glaucoma Suspect PPP should be consulted.[225]

‡ Yearly follow-up if no significant findings; otherwise, follow-up interval based on findings.

Category III: Conditions That Require Intervention

For a patient with ophthalmic or refractive abnormalities, the ophthalmologist prescribes glasses, contact lenses, or other optical devices; treats with medications; arranges for additional evaluation, testing, and follow-up as appropriate; and performs nonsurgical or surgical procedures, including laser surgery, when indicated.

The ophthalmologist should ensure that the patient is informed of relevant examination findings and any need for further evaluation, testing, treatment, or follow-up. Also, relevant ophthalmic findings should be shared with the patient's primary care physician or other specialists, as

appropriate. For a patient with systemic abnormalities, the ophthalmologist may advise further evaluation or referral, as appropriate.

Vision rehabilitation attempts to restore as much functional ability as possible,[226] and patients with reduced visual function benefit from referral for vision rehabilitation and social services (see Vision Rehabilitation PPP).[227] More information on vision rehabilitation, including materials for patients, is available at https://www.aao.org/education/low-vision-and-vision-rehab.

CLINICIAN AND SETTING

Of all health care clinicians, the ophthalmologist, as a physician with full medical training, best combines a thorough understanding of ocular pathology and disease processes; familiarity with systemic disorders that have ocular manifestations; and clinical skills and experience in ocular diagnosis, treatment, and medical decision making. This makes the ophthalmologist the most qualified professional to perform, oversee, and interpret the results of a comprehensive medical eye evaluation. Frequently, and appropriately, specific testing and data collection are conducted by trained personnel working under the ophthalmologist's supervision.

APPENDIX 1. QUALITY OF OPHTHALMIC CARE CORE CRITERIA

Providing quality care
is the physician's foremost ethical obligation, and is
the basis of public trust in physicians.
AMA Board of Trustees, 1986

Quality ophthalmic care is provided in a manner and with the skill that is consistent with the best interests of the patient. The discussion that follows characterizes the core elements of such care.

The ophthalmologist is first and foremost a physician. As such, the ophthalmologist demonstrates compassion and concern for the individual, and utilizes the science and art of medicine to help alleviate patient fear and suffering. The ophthalmologist strives to develop and maintain clinical skills at the highest feasible level, consistent with the needs of patients, through training and continuing education. The ophthalmologist evaluates those skills and medical knowledge in relation to the needs of the patient and responds accordingly. The ophthalmologist also ensures that needy patients receive necessary care directly or through referral to appropriate persons and facilities that will provide such care, and he or she supports activities that promote health and prevent disease and disability.

The ophthalmologist recognizes that disease places patients in a disadvantaged, dependent state. The ophthalmologist respects the dignity and integrity of his or her patients and does not exploit their vulnerability.

Quality ophthalmic care has the following optimal attributes, among others.

- The essence of quality care is a meaningful partnership relationship between patient and physician. The ophthalmologist strives to communicate effectively with his or her patients, listening carefully to their needs and concerns. In turn, the ophthalmologist educates his or her patients about the nature and prognosis of their condition and about proper and appropriate therapeutic modalities. This is to ensure their meaningful participation (appropriate to their unique physical, intellectual, and emotional state) in decisions affecting their management and care, to improve their motivation and compliance with the agreed plan of treatment, and to help alleviate their fears and concerns.
- The ophthalmologist uses his or her best judgment in choosing and timing appropriate diagnostic and therapeutic modalities as well as the frequency of evaluation and follow-up, with due regard to the urgency and nature of the patient's condition and unique needs and desires.
- The ophthalmologist carries out only those procedures for which he or she is adequately trained, experienced, and competent, or, when necessary, is assisted by someone who is, depending on the urgency of the problem and availability and accessibility of alternative providers.
- Patients are assured access to, and continuity of, needed and appropriate ophthalmic care, which can be described as follows.
  - The ophthalmologist treats patients with due regard to timeliness, appropriateness, and his or her own ability to provide such care.
  - The operating ophthalmologist makes adequate provision for appropriate pre- and postoperative patient care.
  - When the ophthalmologist is unavailable for his or her patient, he or she provides appropriate alternate ophthalmic care, with adequate mechanisms for informing patients of the existence of such care and procedures for obtaining it.
  - The ophthalmologist refers patients to other ophthalmologists and eye care providers based on the timeliness and appropriateness of such referral, the patient's needs, the competence and qualifications of the person to whom the referral is made, and access and availability.

  - The ophthalmologist seeks appropriate consultation with due regard to the nature of the ocular or other medical or surgical problem. Consultants are suggested for their skill, competence, and accessibility. They receive as complete and accurate an accounting of the problem as necessary to provide efficient and effective advice or intervention, and in turn they respond in an adequate and timely manner. The ophthalmologist maintains complete and accurate medical records.
  - On appropriate request, the ophthalmologist provides a full and accurate rendering of the patient's records in his or her possession.
  - The ophthalmologist reviews the results of consultations and laboratory tests in a timely and effective manner and takes appropriate actions.
  - The ophthalmologist and those who assist in providing care identify themselves and their profession.
  - For patients whose conditions fail to respond to treatment and for whom further treatment is unavailable, the ophthalmologist provides proper professional support, counseling, rehabilitative and social services, and referral as appropriate and accessible.
- Prior to therapeutic or invasive diagnostic procedures, the ophthalmologist becomes appropriately conversant with the patient's condition by collecting pertinent historical information and performing relevant preoperative examinations. Additionally, he or she enables the patient to reach a fully informed decision by providing an accurate and truthful explanation of the diagnosis; the nature, purpose, risks, benefits, and probability of success of the proposed treatment and of alternative treatment; and the risks and benefits of no treatment.
- The ophthalmologist adopts new technology (e.g., drugs, devices, surgical techniques) in judicious fashion, appropriate to the cost and potential benefit relative to existing alternatives and to its demonstrated safety and efficacy.
- The ophthalmologist enhances the quality of care he or she provides by periodically reviewing and assessing his or her personal performance in relation to established standards, and by revising or altering his or her practices and techniques appropriately.
- The ophthalmologist improves ophthalmic care by communicating to colleagues, through appropriate professional channels, knowledge gained through clinical research and practice. This includes alerting colleagues of instances of unusual or unexpected rates of complications and problems related to new drugs, devices, or procedures.
- The ophthalmologist provides care in suitably staffed and equipped facilities adequate to deal with potential ocular and systemic complications requiring immediate attention.
- The ophthalmologist also provides ophthalmic care in a manner that is cost effective without unacceptably compromising accepted standards of quality.

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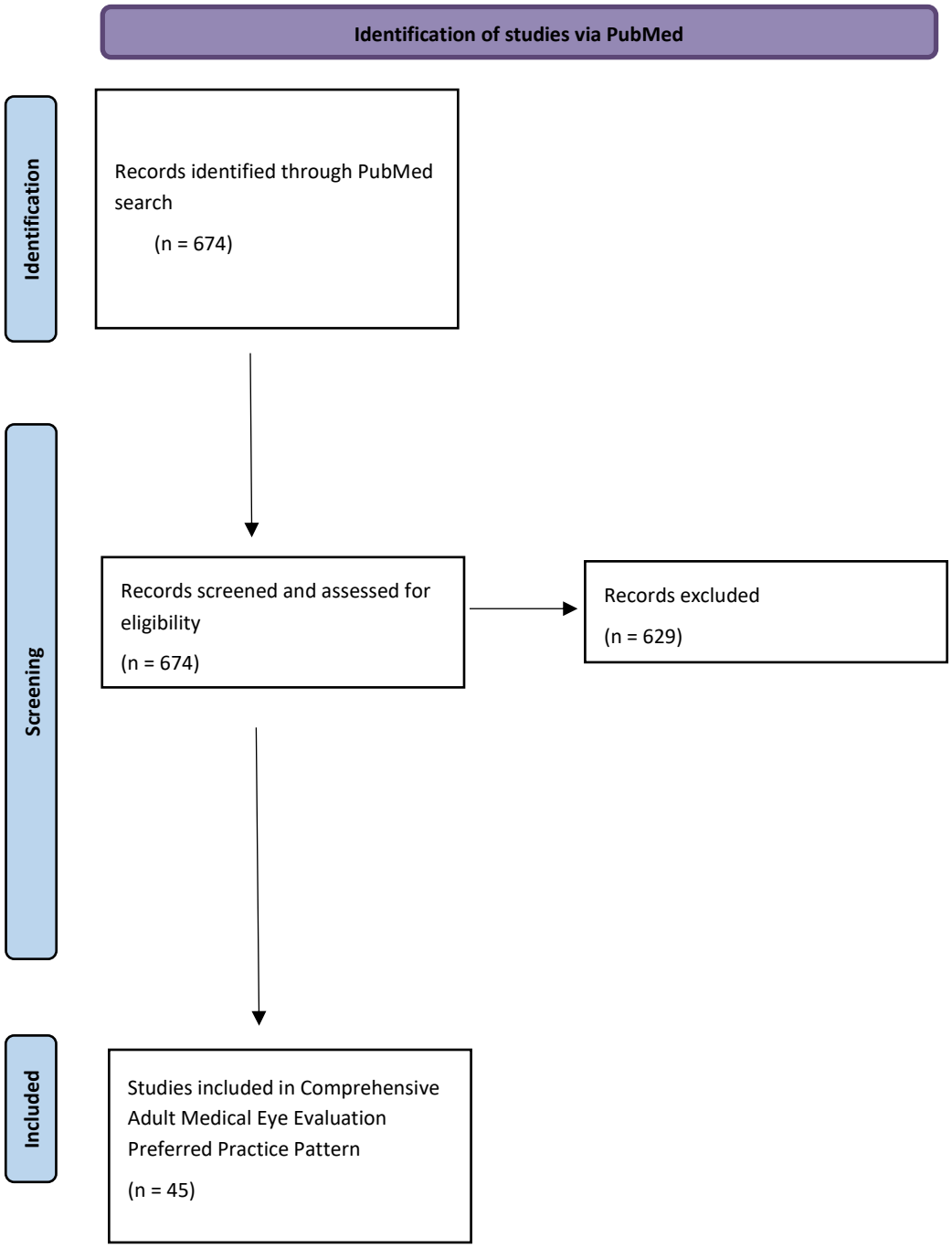
APPENDIX 2. LITERATURE SEARCHES FOR THIS PPP

Literature searches of the PubMed and Cochrane databases were conducted on March 4, 2025; the search strategies are listed below. The searches had added filters for human, English-language randomized controlled trials and systematic reviews and date limiters to capture literature published since June 15, 2020. The committee analyzed 674 studies, of which 45 were included in the PPP.

Activities of Daily Living: "activities of daily living"[mh] AND ("vision disorders"[mh] OR "visual acuity"[mh] OR "visual fields"[mh] OR "visually impaired persons"[mh])

Quality of Life: "quality of life"[mh] AND ("vision disorders"[mh] OR "visual acuity"[mh] OR "visual fields"[mh] OR "visually impaired persons"[mh])

Diagnosis of Vision Disorders: "diagnosis"[mh] AND "vision disorders"[mh]



From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372 n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

RELATED ACADEMY MATERIALS

Basic and Clinical Science Course
Fundamentals and Principles of Ophthalmology (Section 2, 2024-2025)

Clinical Education
Practical Ophthalmology: A Manual for Beginning Residents, 8th ed. (2021)
Ophthalmic Medical Assisting, 7th ed. (2022)

To order any of these products, except for the free materials, please contact the Academy's Customer Service at 866.561.8558 (U.S. only) or 415.561.8540 or www.aao.org/store.

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